3 The influence of the zeros on the step or frequency responses

3.1 Step response

1. Plotting the step response. The MATLAB script assumes $G(s) = \frac{1}{(s+1)(0.5s+1)} = \frac{1}{s+1}\frac{1}{0.5s+1}$ and $G_c(s) = \frac{-s+c}{c(s+1)(0.5s+1)} = \frac{-s+c}{c}\frac{1}{s+1}\frac{1}{0.5s+1}$.

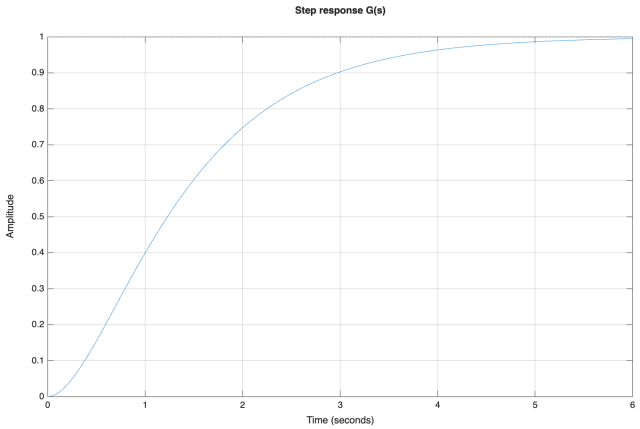


Figure 1: Unit-step response of $G(s)$

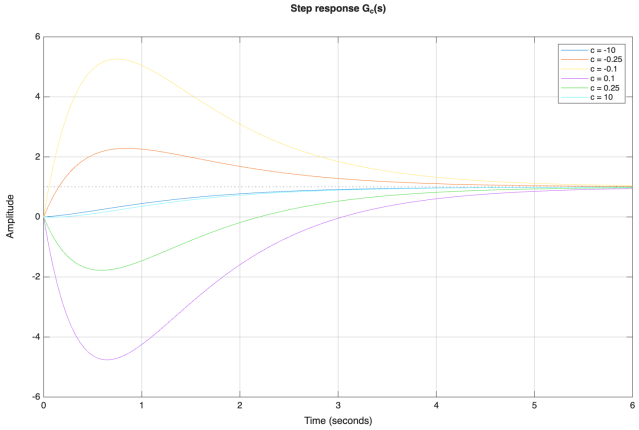


Figure 2: Unit-step responses for different zero locations c.

2. The overshoot and undershoot values are shown in the tables below.

Overshoot (%)	Undershoot (%)
0.00	0.00

Table 1: Overshoot and undershoot of $G(s)$.

c	Overshoot (%)	Undershoot (%)
-10.00	0.00	0.00
-0.25	128.53	0.00
-0.10	426.27	0.00
0.10	0.00	476.19
0.25	0.00	177.75
10.00	0.00	0.83

Table 2: Overshoot and undershoot of $G_c(s)$.

Writing $G(s) = \frac{2}{s^2+3s+2}$ in standard second-order form gives $\omega_n^2 = 2$ and $2\xi\omega_n = 3$, hence $\xi = 3/(2\sqrt{2}) > 1$. This is the second-order case with two real negative poles and no zero. Its unit-step response rises from 0 towards the final value 1, without oscillation, overshoot or undershoot.

For $G_c(s)$ with $c < 0$, the zero adds a temporary positive contribution while the original response is rising. For $c = -0.10$ and $c = -0.25$, it causes an overshoot. A zero close to the origin compared with the poles has a stronger effect, so $c = -0.1$ gives more overshoot than $c = -0.25$. At $c = -10$, the effect is too weak to cause overshoot.

For $G_c(s)$ with $c > 0$, the zero adds a temporary negative contribution, so the response first falls before approaching the final value. This is the undershoot caused by a non minimum phase zero. Moving the zero closer to the origin increases its contribution. $c = 0.1$ produces more undershoot than $c = 0.25$, while $c = 10$ produces only a small undershoot.

3.2 Frequency response

In the graph below we observe the Bode plot of the transfer functions $G(s) = \dfrac{1}{(s+1)(0.5s+1)}$ and $G_c(s) = \dfrac{-s+c}{c\,(s+1)(0.5s+1)}$ for $c = -10,\ -0.25,\ -0.1,\ 0.1,\ 0.25,\ 10$. The zero of G_c sits at $s = c$, and dividing by c keeps the DC gain at 1, so only the zero placement changes between systems.

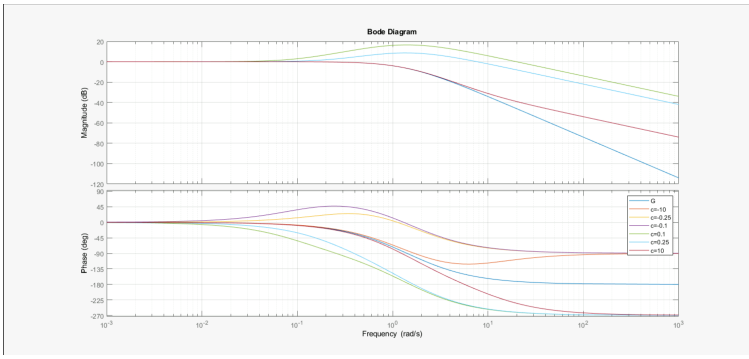


Figure 3: Bode plot of $G(s)$ and the six $G_c(s)$, phase offset adjusted.

In the figure we observe that the magnitude of $G_c(s)$ with the same $|c|$ is the same curve, while each phase curve is distinct. The phases all begin at 0° because of the "Adjust Phase Offset". Left half-plane zeroes ($c < 0$)give a phase lead, so their curves sit above $G(s)$, and Right half-plane zeroes ($c > 0$) give a phase lag, so their curves sit below $G(s)$. This follows from the maths: $G_c(s) = \left(1 - \frac{s}{c}\right) G(s)$, so at $s = j\omega$ the zero adds $\angle\left(1 - \frac{j\omega}{c}\right) = -\arctan(\omega/c)$ to the phase of G. For $c = -\alpha$ this is $+\arctan(\omega/\alpha)$ (lead) and for $c = +\alpha$ it is $-\arctan(\omega/\alpha)$ (lag).

2. Comparison of Gain and Phases of Triples.

Group (α)	Transfer function	$\omega = 0$	$\omega = \alpha$	$\omega \gg \alpha$
1 (0.1)	$G(s)$	0.00	−8.65	−180.00
	$G_{c=0.1}(s)$	0.00	−53.60	−270.00
	$G_{c=-0.1}(s)$	0.00	36.30	−90.00
2 (0.25)	$G(s)$	0.00	−21.10	−180.00
	$G_{c=0.25}(s)$	0.00	−66.20	−270.00
	$G_{c=-0.25}(s)$	0.00	23.80	−90.00
3 (10)	$G(s)$	0.00	−163.00	−180.00
	$G_{c=10}(s)$	0.00	−208.00	−270.00
	$G_{c=-10}(s)$	0.00	−118.00	−90.00

Table 3: Phase (degrees) for each triple.

In the table, at $\omega = 0$ all three have the same phase of about 0°. As frequency increases the phase curves separate. At $\omega = \alpha$, $G_{c<0}$ has gained a lead of approximately 45° on G and $G_{c>0}$ lags G by approximately 45°, so $G_{c>0}$ lags $G_{c<0}$ by 90° (e.g. $\alpha = 10$: −118°, −163°, −208°). For $\omega \gg \alpha$ the gap widens: $G_{c<0}$ leads G by 90° and $G_{c>0}$ lags it by 90°, giving a difference of 180°. In general the gap is $2\arctan(\omega/\alpha)$. The size of α matters too. The poles are at 1 and 2 rad/s, so for $\alpha = 10$ the zero acts well after the poles and the three curves are close in the useful band. For $\alpha = 0.25$ and 0.1 the zero acts first, giving a large gain increase and a large phase difference at low frequency.

The right-half-plane zero satisfies

$$G_{c=\alpha}(s) = G_{c=-\alpha}(s)\frac{\alpha - s}{\alpha + s},$$

where the additional factor has unit magnitude but adds $2\arctan(\omega/\alpha)$ of phase lag. Thus, it gives the same gain but greater phase lag than its left-half-plane counterpart. Of the three systems compared, it

has the greatest phase lag and is therefore non-minimum phase. This zero also causes the undershoot observed in Section 3.1.